

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 02 · Polynomials

With Answers

Questions with full worked solutions

- 37 pages
- Past papers up to January 2026

PURE 2 | TOPIC 02

Polynomials

Your classified route

15

QUESTIONS

117

MARKS

4

FAMILIES

4

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 02 | FAMILY 01

Polynomial Division

WHAT THIS FAMILY ASKS

Recognise the polynomial division demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

19 total marks

T02

02_DIV

02_DIV_CORE

CHAPTER
Polynomials

FAMILY
Polynomial division

VARIANT
core family move

12 marks

WMA12/01 OCTOBER 2025

Question 4

Past-paper question

WMA12-01_2025_Oct_Q04

4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

- (a) When $f(x)$ is divided by $(x - 2)$ the remainder is R and the quotient is $Q(x)$.
- (i) Find $Q(x)$.
- (ii) Find R . (4)
- (b) (i) Use the factor theorem to show that $(x + 4)$ is a factor of $f(x)$.
- (ii) Hence prove, using algebra, that the equation $f(x) = 0$ has only one real solution. (5)
- (c) Find the range of values of x for which $f(x)$ is decreasing. (3)

T02 Polynomials

02_DIV Polynomial division

02_DIV_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 October 2025

WMA12-01_2025_Oct_Q04

1

Divide by x minus two

$$4x^3 + 13x^2 - 10x + 8 = (x - 2)(4x^2 + 21x + 32) + 72.$$

2

State Q and R

$$\text{Hence } Q(x) = 4x^2 + 21x + 32 \text{ and } R = 72.$$

3

Use the factor theorem

$$f(-4) = 4(-4)^3 + 13(-4)^2 - 10(-4) + 8 = -256 + 208 + 40 + 8 = 0.$$

4

Show the factor

Since $f(-4) = 0$, $(x + 4)$ is a factor of $f(x)$.

5

Factor the cubic

$$\text{Dividing by } (x + 4) \text{ gives } f(x) = (x + 4)(4x^2 - 3x + 2).$$

6

Check the quadratic

The discriminant of $4x^2 - 3x + 2$ is $(-3)^2 - 4(4)(2) = 9 - 32 = -23 < 0$, so this quadratic has no real roots.

7

Conclude the real solution

Therefore $f(x) = 0$ has only one real solution, $x = -4$.

8

Differentiate

$$f'(x) = 12x^2 + 26x - 10 = 2(3x - 1)(2x + 5).$$

9

Find where f is decreasing

$f(x)$ is decreasing when $f'(x) < 0$, which is between the roots. Hence $-\frac{5}{2} < x < \frac{1}{3}$.

FINAL ANSWER

WMA12-01_2025_Oct_Q04

(a)(i) $Q(x) = 4x^2 + 21x + 32$. (a)(ii) $R = 72$. (b) **Proven; only real solution is**
 $x = -4$. (c) $-\frac{5}{2} < x < \frac{1}{3}$.

T02

02_DIV

02_DIV_CORE

CHAPTER

Polynomials

FAMILY

Polynomial division

VARIANT

core family move

7 marks

WMA12/01 JANUARY 2026

Question 3

Past-paper question

WMA12-01_2026_Jan_Q03

3. (i) Given that

- $f(x) = 4x^3 + 6x + k$, where k is a constant
- $(x + 2)$ is a factor of $f(x)$

find the remainder when $f(x)$ is divided by $(x - 5)$

(4)

(ii) Find the remainder, R , and the quotient, $Q(x)$, when

$$6x^3 - 15x^2 - 21x + 8$$

is divided by $(2x + 3)$

(3)

T02 Polynomials

02_DIV Polynomial division

02_DIV_CORE core family
move

WORKED SOLUTION

Question 3

WMA12/01 January 2026

WMA12-01_2026_Jan_Q03

1

Use the factor theorem

Since $(x + 2)$ is a factor of $f(x) = 4x^3 + 6x + k$, we have $f(-2) = 0$.

2

Find k

$4(-2)^3 + 6(-2) + k = 0$, so $-32 - 12 + k = 0$ and $k = 44$.

3

Find the required remainder

The remainder on division by $(x - 5)$ is $f(5) = 4(5)^3 + 6(5) + 44 = 574$.

4

Set up the division

Write $6x^3 - 15x^2 - 21x + 8 = (2x + 3)(Ax^2 + Bx + C) + R$.

5

Match coefficients

Comparing coefficients gives $A = 3$, $B = -12$, and $C = \frac{15}{2}$.

6

Find the remainder

Then $(2x + 3)\left(3x^2 - 12x + \frac{15}{2}\right) = 6x^3 - 15x^2 - 21x + \frac{45}{2}$, so $R = 8 - \frac{45}{2} = -\frac{29}{2}$.

FINAL ANSWER

WMA12-01_2026_Jan_Q03

(i) 574. (ii) $Q(x) = 3x^2 - 12x + \frac{15}{2}$, $R = -\frac{29}{2}$.

TOPIC 02 | FAMILY 02

Factor Theorem

WHAT THIS FAMILY ASKS

Recognise the factor theorem demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**7 classified questions**

58 total marks

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

8 marks

WMA12/01 JANUARY 2020

Question 3

Past-paper question

WMA12-01_2020_Jan_Q03

3.

$$f(x) = 6x^3 + 17x^2 + 4x - 12$$

- (a) Use the factor theorem to show that $(2x + 3)$ is a factor of $f(x)$.

(2)

- (b) Hence, using algebra, write $f(x)$ as a product of three linear factors.

(4)

- (c) Solve, for $\frac{\pi}{2} < \theta < \pi$, the equation

$$6 \tan^3 \theta + 17 \tan^2 \theta + 4 \tan \theta - 12 = 0$$

giving your answers to 3 significant figures.

(2)

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 3

WMA12/01 January 2020

WMA12-01_2020_Jan_Q03

1

Use the factor theorem

For $(2x + 3)$, use $x = -\frac{3}{2}$. Substituting gives $f\left(-\frac{3}{2}\right) = 6\left(-\frac{3}{2}\right)^3 + 17\left(-\frac{3}{2}\right)^2 + 4\left(-\frac{3}{2}\right) - 12 = 0$.

2

State the factor result

Since $f\left(-\frac{3}{2}\right) = 0$, $(2x + 3)$ is a factor of $f(x)$.

3

Divide by the known factor

$$6x^3 + 17x^2 + 4x - 12 = (2x + 3)(3x^2 + 4x - 4).$$

4

Factorise the quadratic

$$3x^2 + 4x - 4 = (3x - 2)(x + 2), \text{ so } f(x) = (2x + 3)(3x - 2)(x + 2).$$

5

Use the factorisation for tan theta

Let $t = \tan \theta$. Then $(2t + 3)(3t - 2)(t + 2) = 0$, so $t = -\frac{3}{2}$, $t = \frac{2}{3}$, or $t = -2$.

6

Choose the roots in the interval

For $\frac{\pi}{2} < \theta < \pi$, $\tan \theta$ is negative, so use $\tan \theta = -\frac{3}{2}$ and $\tan \theta = -2$.

7

Find theta

This gives $\theta = \pi - \tan^{-1}\left(\frac{3}{2}\right) = 2.16$ and $\theta = \pi - \tan^{-1}(2) = 2.03$, to 3 significant figures.

FINAL ANSWER

WMA12-01_2020_Jan_Q03

(a) **Proven.** (b) $f(x) = (2x + 3)(3x - 2)(x + 2)$. (c) $\theta = 2.03, 2.16$ **radians.**

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

8 marks

WMA12/01 OCTOBER 2021

Question 4

Past-paper question

WMA12-01_2021_Oct_Q04

4.

$$f(x) = (x^2 - 2)(2x - 3) - 21$$

- (a) State the value of the remainder when $f(x)$ is divided by $(2x - 3)$

(1)

- (b) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$

(2)

- (c) Hence,

- (i) factorise $f(x)$

- (ii) show that the equation $f(x) = 0$ has only one real root.

(5)

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 October 2021

WMA12-01_2021_Oct_Q04

1

Use the remainder theoremDividing by $(2x - 3)$ means use $x = \frac{3}{2}$.

2

Find the remainder

$$f\left(\frac{3}{2}\right) = \left(\frac{9}{4} - 2\right)(0) - 21 = -21.$$

3

Check x equals three

$$f(3) = (9 - 2)(6 - 3) - 21 = 7 \cdot 3 - 21 = 0.$$

4

Use the factor theoremSince $f(3) = 0$, $(x - 3)$ is a factor of $f(x)$.

5

Expand f

$$f(x) = (x^2 - 2)(2x - 3) - 21 = 2x^3 - 3x^2 - 4x - 15.$$

6

FactoriseDividing by $(x - 3)$ gives $2x^2 + 3x + 5$, so $f(x) = (x - 3)(2x^2 + 3x + 5)$.

7

Check the quadratic rootsFor $2x^2 + 3x + 5$, the discriminant is $3^2 - 4(2)(5) = 9 - 40 = -31 < 0$.

8

ConcludeThe quadratic has no real roots, so $f(x) = 0$ has only one real root, $x = 3$.**FINAL ANSWER**

WMA12-01_2021_Oct_Q04

(a) -21 . (b) **Proven.** (c)(i) $f(x) = (x - 3)(2x^2 + 3x + 5)$. (c)(ii) **Only real root:**
 $x = 3$.

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

9 marks

WMA12/01 JANUARY 2023

Question 5

Past-paper question

WMA12-01_2023_Jan_Q05

5.
$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51 \quad (2)$$

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0 \quad (2)$$

(c) Hence find the value of p and the value of q .

(3)

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)g(x) \quad (2)$$

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 5

WMA12/01 January 2023

WMA12-01_2023_Jan_Q05

1

Use the factor theoremSince $(x - 3)$ is a factor, $f(3) = 0$.

2

Substitute x equals three $27 + 9(p + 3) - 3 + q = 0$, so $51 + 9p + q = 0$.

3

Show the first relationTherefore $9p + q = -51$.

4

Use the remainder theoremDividing by $(x + p)$ means substituting $x = -p$. The remainder is $f(-p)$.

5

Substitute x equals negative p $f(-p) = -p^3 + (p + 3)p^2 + p + q = 3p^2 + p + q$.

6

Show the second relationThe remainder is 9, so $3p^2 + p + q = 9$, hence $3p^2 + p + q - 9 = 0$.

7

Eliminate qFrom $9p + q = -51$, $q = -51 - 9p$. Substitute into $3p^2 + p + q - 9 = 0$.

8

Solve for p $3p^2 + p - 51 - 9p - 9 = 0$, so $3p^2 - 8p - 60 = 0$. This gives $p = 6$ or $p = -\frac{10}{3}$.

9

Use p greater than zeroSince $p > 0$, $p = 6$. Then $q = -51 - 9(6) = -105$.

10

Form f with p and q $f(x) = x^3 + 9x^2 - x - 105$.

11

Divide by x minus three $x^3 + 9x^2 - x - 105 = (x - 3)(x^2 + 12x + 35)$, so $g(x) = x^2 + 12x + 35$.**FINAL ANSWER**

WMA12-01_2023_Jan_Q05

(a) **Proven.** (b) **Proven.** (c) $p = 6, q = -105$. (d) $g(x) = x^2 + 12x + 35$.

DR ESLAM AHMED

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

9 marks

WMA12/01 OCTOBER 2023

Question 4

Past-paper question

WMA12-01_2023_Oct_Q04

4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

- (a) show that

$$a + 4b = -56 \quad (2)$$

Given also that when $f(x)$ is divided by $(x - 2)$ the remainder is -25

- (b) find a second simplified equation linking a and b . (2)

- (c) Hence, using algebra and showing your working,

(i) find the value of a and the value of b ,

(ii) fully factorise $f(x)$. (5)

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 October 2023

WMA12-01_2023_Oct_Q04

1

Use the factor theorem

Since $(2x + 1)$ is a factor, $x = -\frac{1}{2}$ is a root, so $f\left(-\frac{1}{2}\right) = 0$.

2

Substitute x equals negative half

$$4\left(-\frac{1}{2}\right)^3 + a\left(-\frac{1}{2}\right)^2 - 29\left(-\frac{1}{2}\right) + b = 0.$$

3

Show the first relation

$-\frac{1}{2} + \frac{a}{4} + \frac{29}{2} + b = 0$. Multiplying by 4 gives $a + 4b + 56 = 0$, so $a + 4b = -56$.

4

Use the remainder theorem

Dividing by $(x - 2)$ gives remainder $f(2)$.

5

Substitute x equals two

$$f(2) = 4(8) + 4a - 58 + b = 4a + b - 26.$$

6

Find the second relation

The remainder is -25 , so $4a + b - 26 = -25$, hence $4a + b = 1$.

7

Solve the simultaneous equations

Solve $a + 4b = -56$ and $4a + b = 1$. Multiplying the second equation by 4 gives $16a + 4b = 4$.

8

Find a and b

Subtracting $a + 4b = -56$ from $16a + 4b = 4$ gives $15a = 60$, so $a = 4$. Then $16 + b = 1$, so $b = -15$.

9

Write f(x)

Therefore $f(x) = 4x^3 + 4x^2 - 29x - 15$.

10

Factorise

Using the known factors $(2x + 1)$ and $(x - 2)$, division gives $f(x) = (2x + 1)(x - 2)(2x + 15)$.

FINAL ANSWER

WMA12-01_2023_Oct_Q04

(a) **Proven.** (b) $4a + b = 1$. (c)(i) $a = 4, b = -15$. (c)(ii) $f(x) = (2x + 1)(x - 2)(2x + 15)$.

DR ESLAM AHMED

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

8 marks

WMA12/01 MAY/JUNE 2024

Question 4

Past-paper question

WMA12-01_2024_MayJune_Q04

4. $f(x) = (x - 2)(2x^2 + 5x + k) + 21$

where k is a constant.

- (a) State the remainder when $f(x)$ is divided by $(x - 2)$ (1)

Given that $(2x - 1)$ is a factor of $f(x)$

- (b) show that $k = 11$ (2)

- (c) Hence

- (i) fully factorise $f(x)$,
(ii) find the number of real solutions of the equation

$$f(x) = 0$$

giving a reason for your answer. (5)

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q04

1

Use the remainder theoremThe remainder on division by $(x - 2)$ is $f(2)$.

2

Find f of two

$$f(2) = (2 - 2)(2(2)^2 + 5(2) + k) + 21 = 21.$$

3

Use the factor theoremSince $(2x - 1)$ is a factor, $x = \frac{1}{2}$ is a root, so $f\left(\frac{1}{2}\right) = 0$.

4

Substitute x equals half

$$\left(\frac{1}{2} - 2\right) \left(2\left(\frac{1}{2}\right)^2 + 5\left(\frac{1}{2}\right) + k\right) + 21 = 0.$$

5

Solve for k

$$-\frac{3}{2}(3 + k) + 21 = 0. \text{ Multiplying by 2 gives } -3(k + 3) + 42 = 0, \text{ so } k = 11.$$

6

Write f with k equals eleven

$$f(x) = (x - 2)(2x^2 + 5x + 11) + 21 = 2x^3 + x^2 + x - 1.$$

7

Factorise

$$2x^3 + x^2 + x - 1 = (2x - 1)(x^2 + x + 1).$$

8

Count real solutions

The quadratic $x^2 + x + 1$ has discriminant $1 - 4 = -3$, so it has no real roots. Therefore $f(x) = 0$ has one real solution.

FINAL ANSWER

WMA12-01_2024_MayJune_Q04

(a) 21. (b) $k = 11$. (c)(i) $(2x - 1)(x^2 + x + 1)$. (c)(ii) **One real solution.**

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

7 marks

WMA12/01 OCTOBER 2024

Question 3

Past-paper question

WMA12-01_2024_Oct_Q03

3.

$$f(x) = 2x^3 - x^2 + Ax + B$$

where A and B are integers.

Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(a) show that

$$3A - B = -118 \quad (2)$$

Given also that $(2x - 5)$ is a factor of $f(x)$,

(b) find the value of A and the value of B . (3)

(c) Hence find the quotient when $f(x)$ is divided by $(x - 7)$ (2)

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 3

WMA12/01 October 2024

WMA12-01_2024_Oct_Q03

1

Use the remainder theoremDividing by $x + 3$ leaves remainder 55, so $f(-3) = 55$.

2

Show the given equation $f(-3) = 2(-3)^3 - (-3)^2 - 3A + B = -63 - 3A + B$. Therefore $-63 - 3A + B = 55$, so $3A - B = -118$.

3

Use the factor theoremSince $2x - 5$ is a factor, $x = \frac{5}{2}$ is a root and $f\left(\frac{5}{2}\right) = 0$.

4

Form a second equation $2\left(\frac{5}{2}\right)^3 - \left(\frac{5}{2}\right)^2 + \frac{5}{2}A + B = 0$, so $25 + \frac{5}{2}A + B = 0$. Multiplying by 2 gives $5A + 2B = -50$.

5

Solve simultaneouslyFrom $3A - B = -118$, $B = 3A + 118$. Substitute into $5A + 2B = -50$ to get $11A = -286$, so $A = -26$ and $B = 40$.

6

Write the polynomial $f(x) = 2x^3 - x^2 - 26x + 40$.

7

Divide by x minus seven $2x^3 - x^2 - 26x + 40 = (x - 7)(2x^2 + 13x + 65) + 495$, so the quotient is $2x^2 + 13x + 65$.**FINAL ANSWER**

WMA12-01_2024_Oct_Q03

(a) **Proven.** (b) $A = -26, B = 40$. (c) $2x^2 + 13x + 65$.

T02

02_FACT

02_FACT_CORE

CHAPTER
Polynomials

FAMILY
Factor theorem

VARIANT
core family move

9 marks

WMA12/01 JANUARY 2025

Question 5

Past-paper question

WMA12-01_2025_Jan_Q05

5.

$$f(x) = 3x^3 + ax^2 - 10x + b$$

where a and b are constants.

Given that $(3x - 4)$ is a factor of $f(x)$,

(a) show that $16a + 9b = 56$

(2)

Given further that when $f(x)$ is divided by $(x - 2)$ the remainder is b ,

(b) find the value of a and the value of b .

(4)

(c) Hence, using algebra, fully factorise $f(x)$.

(3)

T02 Polynomials

02_FACT Factor theorem

02_FACT_CORE core family
move

WORKED SOLUTION

Question 5

WMA12/01 January 2025

WMA12-01_2025_Jan_Q05

1

Use the factor theorem

Since $3x - 4$ is a factor, $x = \frac{4}{3}$ is a root and $f\left(\frac{4}{3}\right) = 0$.

2

Show the given equation

$3\left(\frac{4}{3}\right)^3 + a\left(\frac{4}{3}\right)^2 - 10\left(\frac{4}{3}\right) + b = 0$, so $\frac{64}{9} + \frac{16a}{9} - \frac{120}{9} + b = 0$. Multiplying by 9 gives $16a + 9b = 56$.

3

Use the remainder condition

When dividing by $x - 2$, the remainder is $f(2)$. Given that the remainder is b , $f(2) = b$.

4

Find a

$f(2) = 24 + 4a - 20 + b = 4 + 4a + b$. Thus $4 + 4a + b = b$, so $a = -1$.

5

Find b

Substitute $a = -1$ into $16a + 9b = 56$: $-16 + 9b = 56$, so $b = 8$.

6

Write f of x

$f(x) = 3x^3 - x^2 - 10x + 8$.

7

Find another factor

$f(1) = 3 - 1 - 10 + 8 = 0$, so $(x - 1)$ is a factor.

8

Fully factorise

Dividing by $(x - 1)$ gives $3x^2 + 2x - 8 = (3x - 4)(x + 2)$. Hence $f(x) = (x - 1)(x + 2)(3x - 4)$.

FINAL ANSWER

WMA12-01_2025_Jan_Q05

(a) **Proven.** (b) $a = -1, b = 8$. (c) $f(x) = (x - 1)(x + 2)(3x - 4)$.

TOPIC 02 | FAMILY 03

Remainder Theorem

WHAT THIS FAMILY ASKS

Recognise the remainder theorem demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**5 classified questions**

34 total marks

T02

02_REMA

02_REMA_CORE

CHAPTER
Polynomials

FAMILY
Remainder theorem

VARIANT
core family move

8 marks

WMA12/01 OCTOBER 2019

Question 4

Past-paper question

WMA12-01_2019_Oct_Q04

4. $f(x) = (x - 3)(3x^2 + x + a) - 35$ where a is a constant

(a) State the remainder when $f(x)$ is divided by $(x - 3)$.

(1)

Given $(3x - 2)$ is a factor of $f(x)$,

(b) show that $a = -17$

(2)

(c) Using algebra and showing each step of your working, fully factorise $f(x)$.

(5)

T02 Polynomials

02_REMA Remainder
theorem02_REMA_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 October 2019

WMA12-01_2019_Oct_Q04

1

Use the remainder theorem

When $f(x)$ is divided by $(x - 3)$, the remainder is $f(3)$. Since $f(3) = (3 - 3)(3 \cdot 3^2 + 3 + a) - 35$, the remainder is -35 .

2

Use the given factor

Since $(3x - 2)$ is a factor, $x = \frac{2}{3}$ is a root and $f\left(\frac{2}{3}\right) = 0$.

3

Solve for a

$0 = \left(\frac{2}{3} - 3\right) \left(3 \left(\frac{2}{3}\right)^2 + \frac{2}{3} + a\right) - 35 = -\frac{7}{3}(a + 2) - 35$. Hence $-7(a + 2) = 105$, so $a = -17$.

4

Substitute a into f(x)

$f(x) = (x - 3)(3x^2 + x - 17) - 35 = 3x^3 - 8x^2 - 20x + 16$.

5

Divide by the known factor

Dividing $3x^3 - 8x^2 - 20x + 16$ by $(3x - 2)$ gives $x^2 - 2x - 8$.

6

Factorise fully

$x^2 - 2x - 8 = (x + 2)(x - 4)$, so $f(x) = (3x - 2)(x + 2)(x - 4)$.

FINAL ANSWER

WMA12-01_2019_Oct_Q04

(a) -35 . (b) $a = -17$. (c) $f(x) = (3x - 2)(x + 2)(x - 4)$.

T02

02_REMA

02_REMA_CORE

CHAPTER
Polynomials

FAMILY
Remainder theorem

VARIANT
core family move

6 marks

WMA12/01 JANUARY 2021

Question 1

Past-paper question

WMA12-01_2021_Jan_Q01

1.

$$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$$

where a and b are constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is 4

(a) Show that $a + b = -1$

(2)

When $f(x)$ is divided by $(x - 2)$, the remainder is -23

(b) Find the value of a and the value of b .

(4)

T02 Polynomials

02_REMA Remainder
theorem02_REMA_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 January 2021

WMA12-01_2021_Jan_Q01

1

Use the first remainder

Dividing by $(x + 1)$ means use $x = -1$. Hence $f(-1) = 1 - a - 3 - b + 5 = 3 - a - b$.

2

Show the required relation

The remainder is 4, so $3 - a - b = 4$. Therefore $a + b = -1$.

3

Use the second remainder

Dividing by $(x - 2)$ means use $x = 2$. Hence $f(2) = 16 + 8a - 12 + 2b + 5 = 9 + 8a + 2b$.

4

Form the second equation

The remainder is -23 , so $9 + 8a + 2b = -23$, which gives $4a + b = -16$.

5

Solve simultaneously

Subtract $a + b = -1$ from $4a + b = -16$ to get $3a = -15$, so $a = -5$.

6

Find b

Using $a + b = -1$, $-5 + b = -1$, hence $b = 4$.

FINAL ANSWER

WMA12-01_2021_Jan_Q01

(a) **Proven:** $a + b = -1$. (b) $a = -5$, $b = 4$.

T02

02_REMA

02_REMA_CORE

CHAPTER
Polynomials

FAMILY
Remainder theorem

VARIANT
core family move

8 marks

WMA12/01 JANUARY 2022

Question 5

Past-paper question

WMA12-01_2022_Jan_Q05

5. $f(x) = 3x^3 + Ax^2 + Bx - 10$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(a) show that

$$11A + 9B = 83$$

(3)

Given also that $(3x - 2)$ is a factor of $f(x)$,

(b) find the value of A and the value of B .

(3)

(c) Hence find the quadratic expression $g(x)$ such that

$$f(x) = (3x - 2)g(x)$$

(2)

T02 Polynomials

02_REMA Remainder
theorem02_REMA_CORE core family
move

WORKED SOLUTION

Question 5

WMA12/01 January 2022

WMA12-01_2022_Jan_Q05

1

Use x equals one

$$f(1) = 3 + A + B - 10 = A + B - 7, \text{ so } k = A + B - 7.$$

2

Use x equals negative one

$$f(-1) = -3 + A - B - 10 = A - B - 13.$$

3

Use the second remainder relation

$$\text{Since } f(-1) = -10k, A - B - 13 = -10(A + B - 7).$$

4

Show the relation

$$\text{Rearranging gives } 11A + 9B = 83.$$

5

Use the factor

$$\text{Since } (3x - 2) \text{ is a factor, } f\left(\frac{2}{3}\right) = 0.$$

6

Form the second equation

$$3\left(\frac{2}{3}\right)^3 + A\left(\frac{2}{3}\right)^2 + B\left(\frac{2}{3}\right) - 10 = 0. \text{ Multiplying by 9 gives } 2A + 3B = 41.$$

7

Solve for A and B

$$\text{Solving } 11A + 9B = 83 \text{ with } 2A + 3B = 41 \text{ gives } A = -8 \text{ and } B = 19.$$

8

Divide by the factor

$$\text{With these values, } f(x) = 3x^3 - 8x^2 + 19x - 10 = (3x - 2)(x^2 - 2x + 5).$$

FINAL ANSWER

WMA12-01_2022_Jan_Q05

(a) **Proven:** $11A + 9B = 83$. (b) $A = -8, B = 19$. (c) $g(x) = x^2 - 2x + 5$.

T02

02_REMA

02_REMA_CORE

CHAPTER
Polynomials

FAMILY
Remainder theorem

VARIANT
core family move

3 marks

WMA12/01 JANUARY 2024

Question 1

Past-paper question

WMA12-01_2024_Jan_Q01

1. $f(x) = ax^3 + 3x^2 - 8x + 2$ where a is a constant

Given that when $f(x)$ is divided by $(x - 2)$ the remainder is 3, find the value of a .

(3)

DR ESLAM AHMED

T02 Polynomials

02_REMA Remainder
theorem02_REMA_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 January 2024

WMA12-01_2024_Jan_Q01

1

Use the remainder theoremDividing by $(x - 2)$ means the remainder is $f(2)$.

2

Substitute x equals two

$$f(2) = a(2)^3 + 3(2)^2 - 8(2) + 2 = 8a + 12 - 16 + 2 = 8a - 2.$$

3

Use the given remainderThe remainder is 3, so $8a - 2 = 3$.

4

Solve for a

$$8a = 5, \text{ hence } a = \frac{5}{8}.$$

FINAL ANSWER

$$a = \frac{5}{8}.$$

WMA12-01_2024_Jan_Q01

T02

02_REMA

02_REMA_CORE

CHAPTER
Polynomials

FAMILY
Remainder theorem

VARIANT
core family move

9 marks

WMA12/01 MAY/JUNE 2025

Question 4

Past-paper question

WMA12-01_2025_MayJune_Q04

4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2$$

(2)

Given further that when $f(x)$ is divided by $(2x - 1)$ the remainder is $\frac{7}{4}$

(b) find the value of a and the value of b .

(4)

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$

(3)

T02 Polynomials

02_REMA Remainder
theorem02_REMA_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q04

1

Use the factor theoremSince $x + 3$ is a factor, $f(-3) = 0$.

2

Show the relation $f(-3) = -27a + 9b - 15 - 3 = 0$, so $9b = 27a + 18$ and therefore $b = 3a + 2$.

3

Use the remainder theoremWhen dividing by $2x - 1$, substitute $x = \frac{1}{2}$. The remainder is $\frac{7}{4}$, so $f\left(\frac{1}{2}\right) = \frac{7}{4}$.

4

Form the second equation $\frac{a}{8} + \frac{b}{4} + \frac{5}{2} - 3 = \frac{7}{4}$. Multiplying by 8 gives $a + 2b - 4 = 14$, so $a + 2b = 18$.

5

Solve for a and bSubstitute $b = 3a + 2$ into $a + 2b = 18$: $a + 6a + 4 = 18$, so $a = 2$ and $b = 8$.

6

Write f of x $f(x) = 2x^3 + 8x^2 + 5x - 3$.

7

Divide by x minus two $2x^3 + 8x^2 + 5x - 3 = (x - 2)(2x^2 + 12x + 29) + 55$.

8

State the resultThe quotient is $2x^2 + 12x + 29$ and the remainder is 55.**FINAL ANSWER**

WMA12-01_2025_MayJune_Q04

(a) **Proven.** (b) $a = 2, b = 8$. (c) **Quotient** $2x^2 + 12x + 29$, **remainder** 55.

TOPIC 02 | FAMILY 04

Roots/Factors

WHAT THIS FAMILY ASKS

Recognise the roots/factors demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

1 classified questions

6 total marks

T02

02_ROOT

02_ROOT_CORE

CHAPTER
Polynomials

FAMILY
Roots/factors

VARIANT
core family move

6 marks

WMA12/01 MAY/JUNE 2023

Question 2

Past-paper question

WMA12-01_2023_MayJune_Q02

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.

Given that $(2x - 3)$ is a factor of $f(x)$,

(a) use the factor theorem to show that $a = -3$

(2)

(b) Hence show that the equation $f(x) = 0$ has only one real root.

(4)

T02 Polynomials

02_ROOT Roots/factors

02_ROOT_CORE core family
move

WORKED SOLUTION

Question 2

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q02

1

Use the factor theorem

Since $(2x - 3)$ is a factor, $x = \frac{3}{2}$ is a root, so $f\left(\frac{3}{2}\right) = 0$.

2

Substitute the root

$$4\left(\frac{3}{2}\right)^3 - 8\left(\frac{3}{2}\right)^2 + 5\left(\frac{3}{2}\right) + a = 0.$$

3

Find a

$$13.5 - 18 + 7.5 + a = 0, \text{ so } 3 + a = 0 \text{ and } a = -3.$$

4

Write the polynomial

$$\text{With } a = -3, f(x) = 4x^3 - 8x^2 + 5x - 3.$$

5

Factor using the known factor

$$4x^3 - 8x^2 + 5x - 3 = (2x - 3)(2x^2 - x + 1).$$

6

Check the quadratic

$$\text{The discriminant of } 2x^2 - x + 1 \text{ is } (-1)^2 - 4(2)(1) = 1 - 8 = -7.$$

7

Conclude about real roots

Since the quadratic has negative discriminant, it has no real roots. Therefore $f(x) = 0$ has only the one real root from $2x - 3 = 0$, namely $x = \frac{3}{2}$.

FINAL ANSWER

WMA12-01_2023_MayJune_Q02

$$(a) a = -3. \quad (b) f(x) = 0 \text{ has only one real root, } x = \frac{3}{2}.$$